Notes: Hassan, Hollander, van Lent & Tahoun (QJE, 2019)
Firm-Level Political Risk: Measurement and Effects

Contents

1 Big picture

- **Question.** How can we measure the political risk faced by individual firms, and how does this firm-level political risk affect investment, hiring, volatility, lobbying, and donations?
- **Core measurement idea.** Use earnings-call transcripts. Count how much of the conversation is about political topics in the neighborhood of words related to risk and uncertainty.
- **Main empirical object.** A firm-quarter political-risk measure, $\text{PRisk}_{i,t}$, plus topic-specific measures for eight political topics.
- **Validation.** High-scoring calls are about genuinely political risks; top political bigrams are intuitive; aggregate averages comove with the Baker–Bloom–Davis EPU index; sector averages line up with regulatory and government-dependence measures.
- **Real effects.** Firms with higher political risk show higher implied and realized volatility and lower investment, capex guidance, and employment growth.
- **Political engagement.** High-political-risk firms increase lobbying, campaign donations, number of donation recipients, and hedged donations across parties.
- **Risk versus sentiment.** The results survive controls for political sentiment, generic sentiment, stock returns before calls, and earnings surprises. The paper argues the measure captures second-moment political risk rather than merely bad political news.
- **Main stylized fact.** Most variation in political risk is at the firm level, not at the aggregate or broad sector level.
- **Topic-specific result.** Firms that discuss risks about a specific topic subsequently lobby on that same topic, not simply on politics in general.
- **Takeaway.** Political risk is granular. It reallocates resources, affects firm behavior, and induces firms to spend resources managing politics.

2 Incremental contribution of the paper

- **Firm-level measurement.** Prior work mostly measured aggregate policy uncertainty or exposure to aggregate political shocks. This paper creates a firm-quarter measure of political risk for thousands of firms over time.
- **Use of conference calls.** The paper introduces earnings-call transcripts as a natural source for measuring what firms and analysts actually worry about. Unlike newspapers, calls contain firm-specific discussion by management and market participants.
- **Text method without hand-coded dictionaries.** The method learns political language from political and nonpolitical training libraries and then searches for political language near risk words. This is more flexible than counting a fixed dictionary of terms.
- **Risk, sentiment, and topic decomposition.** The paper separates political risk from political sentiment and separates overall political risk into eight policy-topic components.
- **New evidence on real effects.** The paper links political risk to stock-market volatility, investment, hiring, capex guidance, sales, lobbying, campaign donations, and federal contracts.
- **Granular political risk.** A key contribution is showing that political risk is largely idiosyncratic across firms. This changes how one should model political uncertainty: not only as an aggregate shock, but also as a source of within-sector dispersion and possible misallocation.
- **Bridge between finance, macro, and political economy.** The paper combines text-as-data, firm investment under uncertainty, lobbying and political connections, and granular macro risk into a single empirical framework.

3 Data and measurement

3.1 Conference-call sample

The paper collects transcripts of all earnings conference calls for U.S.-listed firms from Thomson Reuters StreetEvents between 2002 and 2016. The sample contains 178,173 calls for 7,357 firms. Calls include management presentations and analyst Q&A, so they contain both firm disclosures and market participants' questions.

Interpretation: Conference calls are useful because they reveal firm-specific concerns at high frequency. A political issue only enters the measure if it is discussed by the firm or analysts in a setting focused on current and future firm performance.

3.2 Text-based measure of political risk

The paper defines two training libraries: a political library P and a nonpolitical library N. Each is decomposed into adjacent two-word phrases, or bigrams. The algorithm identifies bigrams that are more characteristic of political text than nonpolitical text and then counts these political bigrams when they appear near risk or uncertainty words in an earnings call.

A compact representation is:

$$\text{PRisk}_{i,t} = \frac{1}{B_{i,t}} \sum_{b \in \mathcal{B}_{i,t}} \mathbf{1}\{b \text{ is political}\}\mathbf{1}\{b \text{ appears near risk/uncertainty language}\}\omega_b,$$

where $B_{i,t}$ is the number of bigrams in firm i's call in quarter t, and ω_b is a weight based on the bigram's relative frequency in political versus nonpolitical training text.

Interpretation: The numerator measures politically relevant risk discussion. The denominator normalizes by call length, so the measure is a share of attention rather than a raw word count.

3.3 Political sentiment and nonpolitical risk

The authors build related measures by changing the conditioning words:

- $\text{PSentiment}_{i,t}$ captures whether political language is accompanied by positive or negative sentiment.
- $\text{NPRisk}_{i,t}$ captures nonpolitical risk discussion.
- $\text{Risk}_{i,t}$ captures overall risk discussion regardless of whether it is political.

Interpretation: These companion measures are crucial for falsification. If political risk simply measured bad news, sentiment controls would eliminate the results. If it simply measured generic risk, nonpolitical risk would explain the same political actions.

3.4 Topic-specific political risk

The paper also constructs topic-specific measures:

$$\text{PRisk}^{T}_{i,t}, \qquad T \in \{\text{Economic Policy \& Budget, Environment, Trade, Institutions \& Political Process, Health, Security \& De}$$

Interpretation: Topic scores allow the authors to connect risk discussion to topic-specific lobbying. The key test is whether firms talking about health-care risk later lobby on health care, rather than simply lobbying more on every topic.

3.5 Firm outcomes

The main outcomes include:

- implied and realized stock-return volatility;
- investment, capex guidance, employment growth, and sales growth;
- lobbying expenditures;
- campaign donations, number of political recipients, and hedged donations;
- federal contracts;
- topic-specific lobbying indicators.

Interpretation: The outcome set is deliberately broad. The measure should predict volatility if it captures risk, real-option retrenchment if it captures investment-relevant uncertainty, and lobbying/donations if firms actively manage political exposure.

4 Models and empirical specifications

4.1 Baseline validation regressions

A typical specification is:

$$Y_{i,t} = \alpha_i + \alpha_t + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t},$$

where $Y_{i,t}$ is a volatility or firm-action outcome, and fixed effects vary across specifications.

Interpretation: A positive β for volatility validates that political risk is priced or perceived as uncertainty. A negative β for investment and hiring is consistent with real-options models of uncertainty.

4.2 Firm action regressions

For investment and hiring, the paper estimates relationships of the form:

$$\frac{I_{i,t}}{K_{i,t-1}} \times 100 = \alpha + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + u_{i,t},$$

$$\frac{\Delta emp_{i,t}}{emp_{i,t-1}} \times 100 = \alpha + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + u_{i,t}.$$

Interpretation: The prediction is $\beta < 0$: firms facing higher political risk delay hard-to-reverse investment and hiring decisions.

4.3 Political engagement regressions

For political actions:

$$\log(1 + Lobby_{i,t+1}) = \alpha + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + u_{i,t+1},$$

$$Donation_{i,t+1} = \alpha + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + u_{i,t+1}.$$

Interpretation: The prediction is $\beta > 0$: firms facing political risk spend resources trying to influence or hedge the political process.

4.4 Mean versus variance controls

To distinguish political risk from political news about expected outcomes, the paper estimates:

$$Y_{i,t} = \alpha_i + \alpha_t + \beta \text{PRisk}_{i,t} + \delta_{i,t} + \Gamma' X_{i,t} + u_{i,t}.$$

It also controls for generic sentiment, stock returns before the call, and earnings surprises.

Interpretation: If PRisk were only bad political news, adding sentiment and market-performance controls should greatly weaken β. The paper finds little change.

4.5 Topic-specific lobbying

For topic T, a stylized regression is:

$$\mathbf{1}\{Lobby^T_{i,t+1} > 0\} \times 100 = \alpha_i + \alpha_t + \theta \text{PRisk}^T_{i,t} + \Gamma' X_{i,t} + u_{i,t+1}.$$

Interpretation: A positive θ shows that the text measure has topic-specific economic content: firms exposed to health-policy risk lobby on health, firms exposed to tax-policy risk lobby on tax, and so on.

5 Identification and empirical design

The paper is primarily a measurement and validation paper, not a paper built around one clean randomized shock. The empirical design therefore relies on a cumulative case: the measure is validated against human-readable examples, aggregate benchmarks, sector-level political exposure, volatility, real actions, falsification measures, fixed effects, and topic-specific lobbying.

5.1 Layer 1: construct a measure before studying outcomes

The measurement step is intentionally separated from the outcome regressions. The bigrams are selected from external political and nonpolitical training libraries, not from the outcomes later used in the paper. This reduces mechanical overfitting.

Interpretation: The measure is not constructed to maximize correlation with investment, volatility, or lobbying. It is constructed to identify political-risk language and then tested on economic outcomes.

5.2 Layer 2: face validity from examples and bigrams

Table II shows that the political bigrams are intuitive, and Table III shows that the highest-scoring calls truly discuss political risk. These exercises are not causal tests, but they are essential for measurement validity.

Interpretation: A text measure must first pass a reading test. If high-scoring calls were about generic risk or irrelevant words, subsequent regressions would be uninformative.

5.3 Layer 3: external validation against aggregate and sector benchmarks

Figure I compares the average firm-level measure to the EPU index. Figure II shows election timing. Figure III relates sector averages of the measure to sector exposure to regulation and government spending.

Interpretation: These tests show that the text measure behaves like political risk when aggregated to levels where other proxies exist.

5.4 Layer 4: within-firm and within-sector variation

Many specifications include time, sector, sector-by-time, and firm fixed effects. This matters because the paper's central claim is not only that aggregate political risk exists, but that firm-specific political risk has economic content.

A demanding specification can be written as:

$$Y_{i,t} = \alpha_i + \lambda_{s(i),t} + \beta \text{PRisk}_{i,t} + \Gamma' X_{i,t} + u_{i,t},$$

where $\lambda_{s(i),t}$ are sector-by-time fixed effects.

Interpretation: Identification then compares firms in the same sector at the same time, asking whether the firm that talks more about political risk behaves differently.

5.5 Layer 5: mean versus variance distinction

The paper controls for political sentiment, generic sentiment, stock returns before the call, and earnings surprises. This addresses the concern that PRisk is merely capturing bad political news or bad firm news.

Interpretation: This is one of the central empirical design choices. The conceptual distinction is between news about the mean of future cash flows and news about the variance or risk around those cash flows.

5.6 Layer 6: falsification using nonpolitical risk

The paper constructs NPRisk and overall Risk. Nonpolitical risk should predict lower investment and hiring, but it should not predict political donations, hedging, or lobbying in the same way.

Interpretation: The falsification logic is sharp: generic risk can explain real-option behavior, but political risk should have extra predictive power for political actions.

5.7 Layer 7: topic-specific match between risk and lobbying

The topic-specific analysis asks whether risks about a given topic predict lobbying on that same topic. This is closer to a validation of economic mechanism than an aggregate correlation.

Interpretation: A firm may lobby more for many reasons. But if environmental-risk discussion predicts environmental lobbying specifically, that is strong evidence that the text measure captures meaningful policy exposure.

5.8 Layer 8: budget-crisis quasi-experimental evidence

The paper's closest causal exercise studies terms associated with Obama-era budget crises, such as the debt ceiling, fiscal cliff, and government shutdown. The authors use crisis-specific bigram usage as an instrument for political risk associated with economic policy and budget.

Interpretation: This exercise supports the interpretation that political risks generated by political gridlock can cause firms to lobby on the affected topic. It is narrower than the paper's main measurement exercise, but it strengthens the causal interpretation for one class of political risk.

5.9 Key threats and how the paper addresses them

- **Measurement error.** Addressed through hand-readable examples, intuitive bigrams, aggregate benchmarks, and sector validation.
- **Bad-news confounding.** Addressed with , generic sentiment, pre-call returns, and earnings surprises.
- **Generic uncertainty.** Addressed with NPRisk and overall Risk falsification exercises.
- **Aggregate or sector shocks.** Addressed with time, sector, sector-by-time, and firm fixed effects.
- **Reverse causality.** The paper cannot fully rule out all reverse causality, but the use of leads, next-quarter political actions, and budget-crisis instruments helps.

Interpretation: The paper should be read as building a new measurement infrastructure and then showing that the measure has economic content. The strongest causal claim is in the narrow budget-crisis exercise; the broader results are compelling validation and correlational evidence.

6 Tables and figures

6.1 Table I: Summary Statistics; Table II: Top 120 Political Bigrams Used in Construction of $\mathrm{PRisk}_{i,t}$

Read: Table I summarizes firm-quarter variables, firm-year variables, and topic scores. Table II lists political bigrams that receive high political weights.

Detailed interpretation: Table I shows that PRisk is highly skewed: the mean exceeds the median and the maximum is far above the typical observation. That is economically important because firm-level political risk is concentrated in a right tail of highly exposed firms. Table II validates the text classifier. Bigrams such as "the constitution," "public opinion," "Federal Reserve," and "White House" are recognizable as political language rather than generic business language.

Economic message: These two tables show the two sides of the measurement contribution: the measure has broad firm-quarter coverage, and it is built from terms that visibly correspond to political discourse.

TABLE I
SUMMARY STATISTICS

	Mean	Median	St. dev.	Min	Max	N
Panel A: Firm-quarter						
$\mathrm{PRisk}_{i,t}$ (standardized)	0.70	0.37	1.00	0.00	6.08	176,173
$\mathrm{PSentiment}_{i,t}$ (standardized)	0.90	0.85	1.00	–2.13	3.96	176,173
$\mathrm{Assets}_{i,t}$ (millions)	15,271	1,217	97,502	0.13	3,069,706	173,887
Realized $\mathrm{volatility}_{i,t}$ (standardized)	1.52	1.27	1.00	0.21	8.31	162,153
Implied $\mathrm{volatility}_{i,t}$ (standardized)	2.05	1.82	1.00	0.46	6.31	115,059
Earnings announcement $\mathrm{surprise}_{i,t}$	–0.01	0.00	1.43	–235.83	301.81	161,403
Stock return 7 days prior to earnings $\mathrm{call}_{i,t}$	0.00	0.00	0.02	–0.24	0.40	148,196
$I_{i,t}/K_{i,t-1}$	0.11	0.09	0.11	–0.03	1.07	119,853
$\Delta\mathrm{capexg}_{i,t}/\mathrm{capexg}_{i,t-1}$	0.01	0.00	0.16	–0.44	0.87	22,520
$\Delta\mathrm{sales}_{i,t}/\mathrm{sales}_{i,t-1}$	0.05	0.02	0.35	–0.98	3.46	173,887
Lobby $\mathrm{expense}_{i,t}$ (thousands)	80.08	0.00	381.08	0.00	15,460.00	147,228
Donation $\mathrm{expense}_{i,t}$ (thousands)	5.13	0.00	27.71	0.00	924.50	176,173
# of $\mathrm{recipients}_{i,t}$	2.73	0.00	14.01	0.00	521.00	176,173
$\mathrm{Hedge}_{i,t}$	0.06	0.00	0.24	0.00	1.00	176,173
Federal $\mathrm{contracts}_{i,t}$ (thousands)	3,516	0.00	49,488	0.00	3,841,392	162,124
PRisk Economic Policy & $\mathrm{Budget}_{i,t}$ (standardized)	0.48	0.22	1.00	0.00	64.75	176,173
PRisk $\mathrm{Environment}_{i,t}$ (standardized)	0.33	0.13	1.00	0.00	88.78	176,173
PRisk $\mathrm{Trade}_{i,t}$ (standardized)	0.30	0.10	1.00	0.00	164.55	176,173
PRisk Institutions & Political $\mathrm{Process}_{i,t}$ (standardized)	0.39	0.16	1.00	0.00	71.69	176,173
PRisk $\mathrm{Health}_{i,t}$ (standardized)	0.27	0.10	1.00	0.00	73.02	176,173
PRisk Security & $\mathrm{Defense}_{i,t}$ (standardized)	0.42	0.19	1.00	0.00	123.42	176,173
PRisk Tax $\mathrm{Policy}_{i,t}$ (standardized)	0.37	0.15	1.00	0.00	97.37	176,173
PRisk Technology & $\mathrm{Infrastructure}_{i,t}$ (standardized)	0.41	0.17	1.00	0.00	66.67	176,173

Table I, part 1. Firm-quarter summary statistics and topic-specific political-risk components. Read this table as the baseline scale of the main variables.

TABLE I
(CONTINUED)

	Mean	Median	St. dev.	Min	Max	N
Panel B: Firm-year						
$\mathrm{PRisk}_{i,t}$ (standardized)	0.90	0.59	1.00	0.00	5.97	48,679
$\mathrm{PSentiment}_{i,t}$ (standardized)	1.09	1.05	1.00	–1.90	4.07	48,679
$\Delta\mathrm{emp}_{i,t}/\mathrm{emp}_{i,t-1}$	0.07	0.03	0.30	–0.78	2.50	45,930
Panel C: Firm-topic-quarter						
$\mathrm{PRisk}^{T}_{i,t}$ (standardized)	0.61	0.27	1.00	0.00	6.34	1,177,824
$\mathrm{Lobby}^{T}_{i,t}$ ($\mathbb{1}$)	0.07	0.00	0.25	0.00	1.00	1,177,824

Notes. This table shows the mean, median, standard deviation, minimum, maximum, and number of nonmissing observations of all variables that are used in the subsequent regression analyses. Panels A, B, and C show the relevant statistics for the regression sample at the firm-year, firm-quarter, and firm-topic-quarter unit of analysis, respectively. In Panel A, $PRisk_{i,t}$ is the average for a given firm and quarter of the transcript-based scores of political risk; in Panel B, it is the average for a given firm and year; and in Panel C, $PRisk^{T}_{i,t}$ is the average for a given firm and quarter of the transcript-based scores of topic T. Each of the three are capped at the 99th percentile and standardized by their respective standard deviation. $PSentiment_{i,t}$ is capped at the 1st and 99th percentile and standardized by its standard deviation. Realized $\mathrm{volatility}_{i,t}$ is the standard deviation of 90-day stock holding returns of firm i in quarter t. Implied $\mathrm{volatility}_{i,t}$ is for 90-day at-the-money options of firm i and time t. Both realized and implied volatility are winsorized at the first and last percentile. Stock return 7 days prior to earnings $\mathrm{call}_{i,t}$ is the average stock return for the seven days prior to the earnings call at date t. Earnings announcement $\mathrm{surprise}_{i,t}$ is defined as $\frac{\mathrm{EPS}_{i,t}-\mathrm{EPS}_{i,t-4}}{\mathrm{price}_{i,t}}$, where $\mathrm{EPS}_{i,t}$ is earnings per share (basic) of firm i at time t, and $\mathrm{price}_{i,t}$ is the closing price of quarter t. Capital investment, $\frac{I_{i,t}}{K_{i,t-1}}$, is a measure for capital expenditure and is calculated recursively using a perpetual-inventory method and winsorized at the first and last percentile. Capex guidance, $\frac{\Delta\mathrm{capexg}_{i,t}}{\mathrm{capexg}_{i,t-1}}$, is the quarter-to-quarter percentage change of the capital expenditure guidance about the closest (usually current) fiscal year-end. We allow for a quarter gap if no guidance (about the same fiscal year-end) was given in the preceding quarter and winsorize the resulting variable at the first and last percentile. $\frac{\Delta\mathrm{sales}_{i,t}}{\mathrm{sales}_{i,t-1}}$ is the change in quarter-to-quarter sales over last quarter's value, winsorized at the first and last percentile. Lobby $\mathrm{expense}_{i,t}$ is the total lobby expense during quarter t by firm i. Donation $\mathrm{expense}_{i,t}$ is the sum of all contributions paid to federal candidates in quarter t by firm i. # of $\mathrm{recipients}_{i,t}$ is defined as the total number of recipients of donations made in quarter t by firm i. $\mathrm{Hedge}_{i,t}$ is a dummy variable equal to 1 if donations to Republicans over donations to Democrats are between the 25th and 75th percentile of the sample. Federal $\mathrm{contracts}_{i,t}$ is the net value from all federal contracts (excluding modifications) of firm i in quarter t. Net hiring, $\frac{\Delta\mathrm{emp}_{i,t}}{\mathrm{emp}_{i,t-1}}$, is the change in year-to-year employment over last year's value and is winsorized at the 1st and 99th percentile. Finally, $PRisk^{T}_{i,t}$, where T = {Economic Policy & Budget, Environment, Trade, Institutions & Political Process, Health, Security & Defense, Tax policy, Technology & Infrastructure}, are the separate topic scores, capped at the 99th percentile and standardized by their respective standard deviation. All variables are restricted to the set of observations of the largest regression sample that is reported in any of the subsequent tables.

Table I, part 2. Additional firm-year and political-action variables. This part shows the scale of lobbying, donations, federal contracts, employment, and Compustat outcomes.

TABLE II

TOP 120 POLITICAL BIGRAMS USED IN CONSTRUCTION OF $PRisk_{i,t}$

Bigram	$\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}} \times 10^5$	Frequency	Bigram	$\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}} \times 10^5$	Frequency
the constitution	201.15	9	governor and	26.79	11
the states	134.29	203	government the	26.39	56
public opinion	119.05	4	this election	25.98	26
interest groups	118.46	8	political party	25.80	5
of government	115.53	316	American political	25.80	2
the GOP	102.22	1	politics of	25.80	5
in Congress	78.00	107	White House	25.80	21
national government	68.03	7	the politics	25.80	31
social policy	62.16	1	general election	25.22	30
the civil	60.99	64	and political	25.22	985
elected officials	60.40	3	policy is	25.22	135
politics is	53.95	7	the islamic	25.04	1
political parties	51.61	3	Federal Reserve	24.63	119
office of	51.02	58	judicial review	24.04	6
the political	51.02	1,091	vote for	23.46	6
interest group	48.09	1	limits on	23.46	53
the bureaucracy	48.09	1	the FAA	23.28	22
and Senate	46.33	19	the presidency	22.87	2
government and	44.57	325	shall not	22.87	4
for governor	41.48	2	the nation	22.87	52
executive branch	40.46	3	constitution and	22.87	3
support for	39.88	147	Senate and	22.87	28
the EPA	39.15	139	the VA	22.65	77
in government	38.70	209	of citizens	22.28	12
Congress to	36.95	19	any state	22.28	7
political process	36.36	18	the electoral	22.28	5
care reform	35.77	106	a president	21.70	6
government in	35.19	77	the governments	21.70	201
due process	35.19	6	clause of	21.11	1
President Obama	34.60	7	and Congress	21.11	7
and social	34.60	140	the partys	21.11	1
first amendment	34.01	1	the Taliban	20.64	1
Congress the	34.01	9	a yes	20.64	12
the Republican	33.43	10	other nations	20.53	1
Tea Party	33.43	1	passed by	20.53	13
the legislative	33.43	92	states or	20.53	40
of civil	32.84	14	free market	20.53	29
court has	32.84	30	that Congress	20.53	30
groups and	32.25	109	national and	20.53	194
struck down	31.67	3	most Americans	19.94	2
shall have	31.67	7	of religion	19.94	1
civil war	31.67	8	powers and	19.94	3
the Congress	31.67	50	a government	19.94	92

Table II, part 1. First portion of the top political bigrams used to construct $\mathrm{PRisk}_{i,t}$*. The list is a face-validity check for the text-classification procedure.*

TABLE II

(CONTINUED)

Bigram	$\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}} \times 10^5$	Frequency	Bigram	$\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}} \times 10^5$	Frequency
he constitutional	29.91	9	politics and	19.94	22
uled that	29.32	15	the South	19.94	406
he presidential	29.32	121	government is	19.94	235
f representatives	28.74	10	yes vote	19.39	1
olicy goals	28.15	2	to enact	19.35	6
frican Americans	28.15	2	political system	19.35	6
conomic policy	28.15	15	proposed by	19.35	25
f social	28.15	31	the legislature	19.35	32
political	28.15	121	the campaign	19.35	41
f speech	27.56	1	federal bureaucracy	18.77	3
ivil service	27.56	2	and party	18.77	2
overnment policy	27.56	52	governor in	18.76	1
ederal courts	27.56	1	state the	18.26	35
rgued that	26.98	8	executive privilege	18.18	1
he democratic	26.98	7	of politics	18.18	4

Table II, part 2. Continuation of the top political bigrams. The frequency columns show that highly weighted words are not necessarily the most common words; informativeness differs from frequency.

6.2 Table III: Transcript Excerpts with Highest $\mathrm{PRisk}_{i,t}$

Read: Table III displays transcript excerpts from calls with the highest political-risk scores.

Detailed interpretation: The excerpts show that the algorithm identifies genuine political risk: smoking bans, regulation, government funding, ballot initiatives, federal policy, and similar risks. This is an important qualitative validation because it helps rule out the possibility that the measure is driven by accidental word proximity.

Economic message: The table connects the machine measure to human interpretation. It shows that PRisk is not just a statistical artifact: it captures the kinds of concerns managers actually discuss with analysts.

TABLE III

TRANSCRIPT EXCERPTS WITH HIGHEST $PRisk_{i,t}$

Firm name	Call date	$PRisk_{i,t}$ (standardized)	Discussion of political risks associated with:	Text surrounding bigram with highest weight ($\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}}$)
Nevada Gold Casinos Inc.	10-Sep-2008	51.94	• impact of statewide smoking ban on revenues; • ballot initiative to amend the constitution to remove caps on bets; • EPA determinations concerning project development.	gaming industry is currently **supporting a ballot** initiative to **amend the constitution to authorize an** increase in the —BET— limits allow additional
Axis Capital Holdings Limited	9-Feb-2010	48.70	• exposure of insurance portfolio to political risk in Spain, Portugal, Greece, Ukraine, and Kazakhstan.	accident year ratios the combined ratios we have talked about **the political** —RISK— business particularly really shouldnt be looked at on a
Female Health	10-Feb-2009	44.17	• developments regarding USAID, a major customer; • FDA approval of company products; • Senate vote on stimulus funding and government funding of AIDS/HIV prevention; • restrictions on funding of organizations that permit abortion.	market acceptance the economic and business environment and the impact **of government pressures** currency —RISKS— capacity efficiency and supply **constraints and** other
Employers Holdings Inc.	01-May-2014	43.81	• passage of California Senate Bill on workers' compensation.	of —HAZARD— **groups but** as you start moving it around **the states** you can have an impact Robert Paun Sidoti company analyst

Table III, part 1. Highest-scoring transcripts and the political risks discussed.

TABLE III

(CONTINUED)

Firm name	Call date	$PRisk_{i,t}$ (standardized)	Discussion of political risks associated with:	Text surrounding bigram with highest weight ($\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}}$)
National Mentor Holdings, Inc.	12-Feb-2010	42.55	• state and federal budgets; • federal stimulus package; • funding of Medicaid.	governments **both President Obama's budget proposal and separate legislation** —PENDING— **in Congress would provide** funding to continue **the Medicaid** stimulus for **another**
Applied Energetics, Inc.	11-May-2009	41.12	• collaboration with Pentagon to develop technology to counter IED/roadside bombs; • funding of weapons programs.	of products and the —UNCERTAINTY— of the timing **and magnitude of government funding and** customer orders **dependence on** sales **to government** customers
Calian Group Ltd	9-Feb-2011	41.05	• impact of revenues of government cost cutting initiatives.	sure Benoit Poirier Desjardins Securities analyst okay and in terms **of government** cost cutting initiatives is there any —RISK— of missing consensus
Insurance Australia Group Ltd	23-Feb-2012	38.70	• Australian election for prime minister; • likelihood of carbon tax introduction.	leadership I just wondered if you **had concerns** about how **the political** —INSTABILITY— might affect **policies that** have **ramifications for** the industry
FPIC Insurance Group, Inc.	30-Oct-2008	38.69	• impact of the composition of Congress on the likelihood of tort reform; • Florida state politics.	a —CHANCE— for national **tort reform and** I don't see **the constitution of** Congress **changing in** such a way after **this election**
BankFinancial Corp	4-Nov-2008	38.33	• TARP and CPP programs; • developments in Freddie Mac; • consequences of a change in administration and party in power.	was **an accurate** metaphor and really given all the —UNCERTAINTIES— **of government involvement** in operations and business activities and given the capital

Table III, part 2. Continuation of transcript examples showing political-risk conversations.

TABLE III

(CONTINUED)

Firm name	Call date	$PRisk_{i,t}$ (standardized)	Discussion of political risks associated with:	Text surrounding bigram with highest weight ($\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}}$)
Nanogen, Inc.	8-Aug-2007	37.20	• FDA approval of company products.	**a dip** in revenues during q related to the —UNCERTAINTY— **of government approval for** the phase **funding of** the cdc contract additionally
World Acceptance Corporation	25-Jul-2006	36.90	• impact of legislation in Texas and other states.	management analyst I wanted to followup on the regulatory **front the states** that you had **mentioned the** —POSSIBILITY— of some positive **legislation**
United Refining Company	23-Jul-2010	35.32	• effect of government tax refund on bottom line; • state funding of infrastructure projects and the associated demand for asphalt products.	shape on asphalt **the funding is** very —IFFY— in all **the states so** and the private work is very slow operator operator
Magellan Health Services	29-Jul-2010	35.26	• actions of state Medicaid administrators and insurance regulators; • state procurement of healthcare reform and federal regulations; • state gubernatorial elections; • Affordable Care Act.	future so this is a time of quite —UNCERTAINTY— for **the states** they are not **sure what** the fmap will be if
Piraeus Bank SA	19-Mar-2015	34.45	• political situation in Greece; • consequences of elections on bank deposits; • relations between EU and Greece, politics of Greece leaving the Euro	that this time around the process or the impact of **the political —uncertainty—** has been a bit more subdued than last time

Table III, part 3. Additional transcript excerpts. The examples help identify true positives and the few false positives.

TABLE III

(CONTINUED)

Firm name	Call date	$PRisk_{i,t}$ (standardized)	Discussion of political risks associated with:	Text surrounding bigram with highest weight ($\frac{f_{b,\mathbb{P}}}{B_{\mathbb{P}}}$)
Piedmont Natural Gas	9-Jun-2009	34.39	—	**your point** as you will recall in all three of **the states** that we have serve Jim we are —EXPOSED— only to
Platinum Underwriters Holdings Ltd	18-Feb-2010	33.21	• politics and government decision-making in Kazakhstan and Ukraine; • China's ability to fulfill lending commitments.	we have had historically had a very small participation in **the political** —RISK— market backing only a couple of players **parties that**
Transcontinental Inc.	14-Sep-2006	31.81	• tax reform in Quebec.	magazines when you look at exports that we do to **the states no** —DOUBT— that is affecting the top and the bottom
Hemisphere Media Group Inc.	12-Aug-2014	31.70	• restructuring of government debt in Puerto Rico.	I think largely a result of the —UNCERTAINTY— regarding restructuring **of government** debt and the general overhang on **the weak economy in**
Pointer Telocation Ltd	30-May-2012	31.27	• political conditions in Israel.	anticipated such —RISKS— and —UNCERTAINTIES— include **a dependence on** economic **and political conditions in Israel the** impact of competition supply constraints as

Notes. This table lists transcripts sorted on $PRisk_{i,t}$ together with their associated firm name, earnings call date, $PRisk_{i,t}$ (standardized), a summary of relevant discussions of political risks in the transcript, and the text surrounding the bigram that has received the highest weight in the transcript. Bigrams for which $b \in \mathbb{P} \backslash \mathbb{N}$ are marked bold; the bigram that received the highest weight is precisely in the middle of the text excerpt. A synonym of "risk" or "uncertainty" is written in small caps and surrounded by dashes. $PRisk_{i,t}$ is standardized by its standard deviation, but not capped because they are in the 99th percentile. Duplicate firms are removed from this top list.

Table III, part 4. Final excerpt panel and notes. The table clarifies the distinction between political risk and generic risk language.

6.3 Figure I: Variation in $\mathrm{PRisk}_{i,t}$ over Time and Correlation with EPU; Figure II: Variation in $\mathrm{PRisk}_{i,t}$ around Federal Elections

Read: Figure I plots the time average of firm-level political risk against the news-based EPU index. Figure II estimates election-quarter effects.

Detailed interpretation: Figure I shows strong comovement between average firm-level PRisk and aggregate EPU. This validates the aggregate time-series component of the measure. Figure II shows that political risk rises around federal elections even after firm fixed effects and firm-size controls. This election pattern is exactly what one would expect if the measure captures political exposure.

Economic message: These figures show that when the measure is aggregated, it recovers familiar aggregate political-risk patterns. But the rest of the paper shows that aggregation hides most of the interesting firm-level variation.

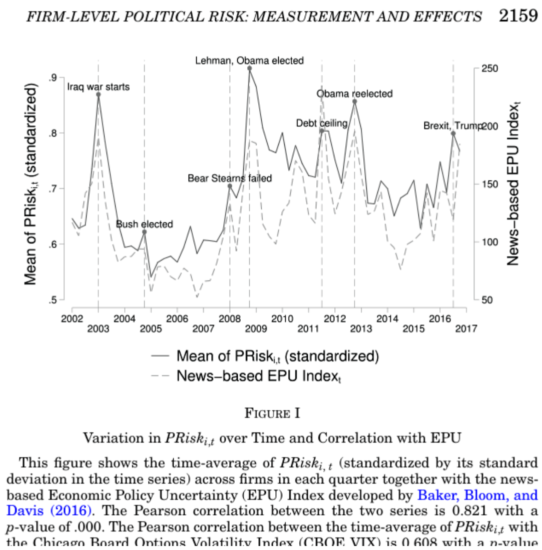

Figure I. Average firm-level political risk and EPU. The high correlation supports aggregate validity.

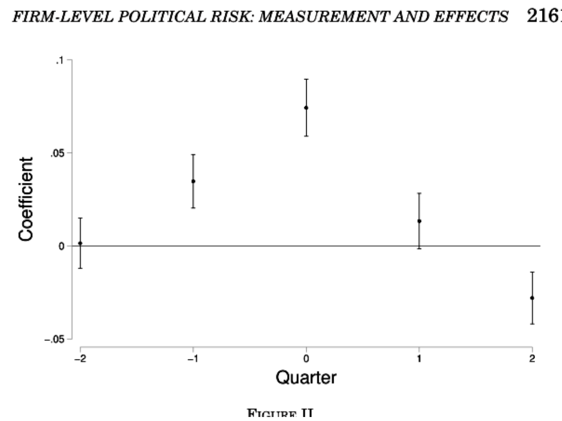

Figure II. Federal election event study. Political-risk discussion rises around election quarters.

6.4 Figure III: $\mathrm{PRisk}_{i,t}$ and Sector Exposure to Politics; Table IV: Validation: Implied and Realized Volatility

Read: Figure III relates sector-average political risk to sector measures of regulation and government spending. Table IV relates PRisk to implied and realized volatility.

Detailed interpretation: Figure III validates the cross-sectional content of the measure: sectors more exposed to regulatory language or government demand have higher measured political risk. Table IV shows that political-risk discussion predicts higher firm volatility. This is a key validation because volatility should react to risk, not simply to political vocabulary.

Economic message: The figure says the measure is high where political exposure should be high; the table says markets also treat these conversations as risk-relevant.

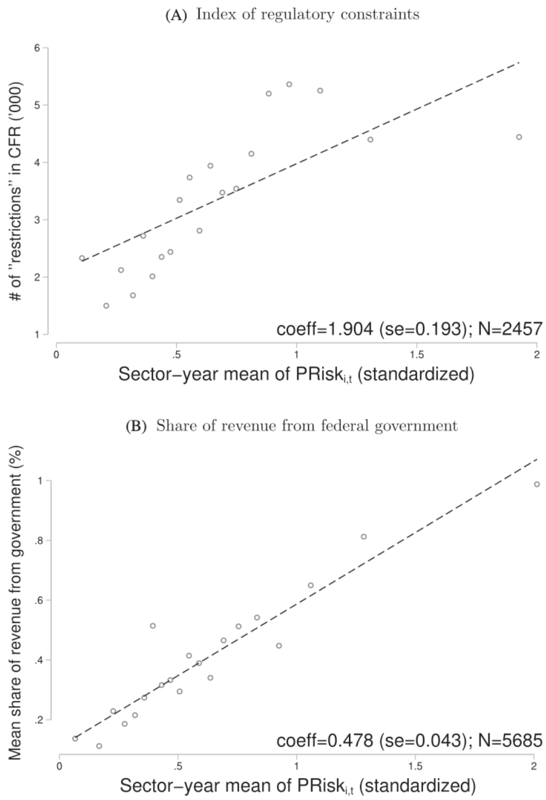

FIGURE III

$PRisk_{i,t}$ and Sector Exposure to Politics

Figure III. Sector-level political risk versus regulatory exposure and government-spending exposure.

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TABLE IV

VALIDATION: IMPLIED AND REALIZED VOLATILITY

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Implied volatility$_{i,t}$ (standardized)						
PRisk$_{i,t}$ (standardized)	0.056***	0.034***	0.033***	0.025***	0.013***	0.016**
	(0.006)	(0.006)	(0.006)	(0.005)	(0.003)	(0.006)
Mean of PRisk$_{i,t}$ (standardized)		0.262***				
		(0.004)				
R^2	0.214	0.275	0.394	0.451	0.711	0.783
N	115,059	115,059	115,059	115,059	115,059	18,060
Panel B: Realized volatility$_{i,t}$ (standardized)						
PRisk$_{i,t}$ (standardized)	0.048***	0.023***	0.020***	0.020***	0.014***	0.013**
	(0.005)	(0.004)	(0.004)	(0.004)	(0.002)	(0.006)
Mean of PRisk$_{i,t}$ (standardized)		0.295***				
		(0.004)				
R^2	0.140	0.224	0.406	0.438	0.621	0.709
N	162,153	162,153	162,153	162,153	162,153	20,816
Time FE	no	no	yes	yes	yes	yes
Sector FE	no	no	no	yes	n/a	n/a
Firm FE	no	no	no	no	yes	yes
CEO FE	no	no	no	no	no	yes

Notes. This table shows the results from regressions with realized and implied volatility as the dependent variable in Panels A and B, respectively. Realized volatility$_{i,t}$ is the standard deviation of 90-day stock holding returns of firm i in quarter t and is winsorized at the first and last percentile. Implied volatility$_{i,t}$ is for 90-day at-the-money options of firm i and time t and is also winsorized at the first and last percentile. $PRisk_{i,t}$ is our measure for firm-level political risk. All regressions control for the log of firm assets. Realized volatility$_{i,t}$, implied volatility$_{i,t}$, and $PRisk_{i,t}$ are standardized by their respective standard deviation. The regression sample in the last column is based on the first quarter of each year due to the annual frequency of CEO information. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table IV. PRisk *predicts implied and realized volatility, supporting the interpretation that the measure captures risk.*

6.5 Table V: Managing Political Risk; Table VI: Mean versus Variance of Political Shocks

Read: Table V studies investment, capex guidance, employment, sales, donations, lobbying, and heterogeneous responses by firm size. Table VI adds controls for sentiment and other first-moment news.

Detailed interpretation: Table V is central. A one-standard-deviation increase in political risk is associated with lower investment, lower capex guidance, and lower employment growth, while sales are much less affected. This is consistent with real-options models: firms delay costly-to-reverse decisions more than they adjust short-run sales. The same table shows that firms increase donations and lobbying after political-risk exposure. Table VI shows that these patterns survive political sentiment, generic sentiment, returns before the call, and earnings surprises, supporting the risk-versus-mean distinction.

Economic message: Firms do not merely talk about political risk; they change behavior. They retrench real activity and increase political engagement. The sentiment controls show that the result is not just bad political news.

TABLE V

MANAGING POLITICAL RISK

	(1)	(2)	(3)	(4)
Panel A	$\frac{I_{i,t}}{K_{i,t-1}} * 100$	$\frac{\Delta \text{capexg}_{i,t}}{\text{capexg}_{i,t-1}} * 100$	$\frac{\Delta \text{emp}_{i,t}}{\text{emp}_{i,t-1}} * 100$	$\frac{\Delta \text{sales}_{i,t}}{\text{sales}_{i,t-1}} * 100$
$PRisk_{i,t}$ (standardized)	−0.159***	−0.338***	−0.769***	−0.075
	(0.041)	(0.120)	(0.155)	(0.094)
R^2	0.035	0.041	0.024	0.016
N	119,853	22,520	45,930	173,887
Panel B	Log(1+$ donations$_{i,t+1}$)	# of recipients$_{i,t+1}$	Hedge$_{i,t+1}$	Log(1+$ lobby$_{i,t+1}$)
$PRisk_{i,t}$ (standardized)	0.087***	0.462***	0.007***	0.186***
	(0.018)	(0.118)	(0.001)	(0.027)
R^2	0.250	0.147	0.140	0.268
N	176,173	176,173	176,173	147,228
Panel C	$\frac{I_{i,t}}{K_{i,t-1}} * 100$	$\frac{\Delta \text{emp}_{i,t}}{\text{emp}_{i,t-1}} * 100$	Log(1+$ donations$_{i,t+1}$)	Log(1+$ lobby$_{i,t+1}$)
$PRisk_{i,t}$ (standardized)	−0.223***	−1.064***	0.025	0.168***
	(0.059)	(0.230)	(0.016)	(0.032)
$PRisk_{i,t} \times \mathbb{1}\{\text{assets}_{i,t} > \text{median assets}\}$	0.149*	0.620**	0.154***	0.085
	(0.081)	(0.289)	(0.039)	(0.056)
N	119,853	45,930	176,173	147,228
Time FE	yes	yes	yes	yes
Sector FE	yes	yes	yes	yes

Notes. Panel A shows the results from regressions of capital investment (column [1]), capital expenditure guidance (column [2]), net hiring (column [3]), and net sales (column [4]) on $PRisk_{i,t}$. Capital investment, $\frac{I_{i,t}}{K_{i,t-1}} * 100$, is calculated recursively using a perpetual-inventory method. Capex guidance, $\frac{\Delta \text{capexg}_{i,t}}{\text{capexg}_{i,t-1}}$, is the quarter-to-quarter percentage change of the capital expenditure guidance about the closest (usually current) fiscal year-end. We allow for a quarter gap if no guidance (about the same fiscal year-end) was given in the preceding quarter. Net hiring, $\frac{\Delta \text{emp}_{i,t}}{\text{emp}_{i,t-1}} * 100$, is the change in year-to-year employment over last year's value. Net sales is defined similarly on quarterly data. Capital investment, net hiring, capital expenditure guidance, and net sales are all winsorized at the first and last percentile. Panel B shows the results of regressions of lobbying and donation activity by firms on $PRisk_{i,t}$. Log(1+$ donations$_{i,t+1}$) (column [1]) is the log of 1 plus the sum of all contributions paid to federal candidates; # of recipients$_{i,t+1}$ (column [2]) is defined as

Table V. Managing political risk. Higher political risk predicts lower investment/hiring and more political activity.

TABLE VI

MEAN VERSUS VARIANCE OF POLITICAL SHOCKS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A	$\frac{I_{i,t}}{K_{i,t-1}} * 100$				$\frac{\Delta \text{emp}_{i,t}}{\text{emp}_{i,t-1}} * 100$			
$PRisk_{i,t}$ (standardized)	−0.159***	−0.145***	−0.120***	−0.157***	−0.769***	−0.683***	−0.534***	−0.622***
	(0.041)	(0.041)	(0.041)	(0.046)	(0.155)	(0.156)	(0.156)	(0.163)
$PSentiment_{i,t}$ (standardized)		0.216***				1.181***		
		(0.043)				(0.155)		
Sentiment$_{i,t}$ (standardized)			0.454***				2.252***	
			(0.048)				(0.161)	
Mean stock return 7 days prior$_{i,t}$ (%)				0.025				0.319*
				(0.022)				(0.166)
Earnings announcement surprise$_{i,t}$				0.058*				0.024***
				(0.032)				(0.005)
R^2	0.035	0.035	0.036	0.037	0.024	0.026	0.029	0.026
N	119,853	119,853	119,853	100,661	45,930	45,930	45,930	41,327
Panel B	Log(1+$ lobby$_{i,t+1}$)				Log(1+$ donations$_{i,t+1}$)			
$PRisk_{i,t}$ (standardized)	0.186***	0.199***	0.204***	0.217***	0.087***	0.094***	0.097***	0.100***
	(0.027)	(0.027)	(0.027)	(0.031)	(0.018)	(0.018)	(0.018)	(0.020)
$PSentiment_{i,t}$ (standardized)		0.203***				0.117***		
		(0.032)				(0.022)		
Sentiment$_{i,t}$ (standardized)			0.203***				0.115***	
			(0.037)				(0.026)	
Mean stock return 7 days prior$_{i,t}$ (%)				0.028***				0.012***
				(0.007)				(0.004)

Table VI, part 1. Mean-versus-variance controls for investment and employment. The PRisk *coefficient remains negative after sentiment controls.*

TABLE VI

(CONTINUED)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Earnings announcement surprise$_{i,t}$				−0.007				−0.003
				(0.007)				(0.004)
R^2	0.268	0.269	0.269	0.291	0.250	0.251	0.251	0.282
N	147,228	147,228	147,228	121,650	176,173	176,173	176,173	147,521
Panel C	# of recipients$_{i,t+1}$				Hedge$_{i,t+1}$			
$PRisk_{i,t}$ (standardized)	0.462***	0.491***	0.509***	0.512***	0.007***	0.007***	0.007***	0.008***
	(0.118)	(0.121)	(0.121)	(0.136)	(0.001)	(0.001)	(0.001)	(0.001)
$PSentiment_{i,t}$ (standardized)		0.474***				0.008***		
		(0.100)				(0.001)		
Sentiment$_{i,t}$ (standardized)			0.541***				0.007***	
			(0.131)				(0.002)	
Mean stock return 7 days prior$_{i,t}$ (%)				0.032**				0.001**
				(0.013)				(0.000)
Earnings announcement surprise$_{i,t}$				0.011				−0.000
				(0.013)				(0.000)
R^2	0.147	0.148	0.149	0.172	0.140	0.141	0.141	0.158
N	176,173	176,173	176,173	147,521	176,173	176,173	176,173	147,521

Notes. In all regressions, $PRisk_{i,t}$, $PSentiment_{i,t}$, and Sentiment$_{i,t}$ are standardized by their standard deviation. Mean stock return 7 days prior$_{i,t}$ (%) is the average stock return for the seven days prior to the earnings call of firm i at date t. Earnings announcement surprise$_{i,t}$ is defined as $\frac{\text{EPS}_{i,t} - \text{EPS}_{i,t-4}}{\text{price}_{i,t}}$, where EPS$_{i,t}$ is earnings per share (basic) of firm i at time t, and price$_{i,t}$ is the closing price of quarter t. The remaining variables are defined as in the preceding tables. All specifications control for the log of firm assets, sector, and time fixed effects. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table VI, part 2. Mean-versus-variance controls for political-action outcomes. Results remain strong after first-moment controls.

TABLE VII

FALSIFICATION EXERCISE: POLITICAL RISK, NONPOLITICAL RISK, AND OVERALL RISK

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A	$\frac{I_{i,t}}{K_{i,t-1}} * 100$			$\frac{\Delta \text{emp}_{i,t}}{\text{emp}_{i,t-1}} * 100$		
$PRisk_{i,t}$ (standardized)	−0.145***	−0.085**	−0.075*	−0.683***	−0.441***	−0.402**
	(0.041)	(0.042)	(0.045)	(0.156)	(0.162)	(0.182)
$NPRisk_{i,t}$ (standardized)		−0.255***			−0.854***	
		(0.043)			(0.166)	
$Risk_{i,t}$ (standardized)			−0.136**			−0.509**
			(0.059)			(0.209)
R^2	0.035	0.036	0.035	0.026	0.027	0.026
N	119,853	119,853	119,853	45,930	45,930	45,930
Panel B	Log(1+$ lobby$_{i,t+1}$)			Log(1+$ donations$_{i,t+1}$)		
$PRisk_{i,t}$ (standardized)	0.199***	0.204***	0.212***	0.094***	0.095***	0.108***
	(0.027)	(0.027)	(0.028)	(0.018)	(0.018)	(0.019)
$NPRisk_{i,t}$ (standardized)		−0.023			−0.004	
		(0.022)			(0.015)	
$Risk_{i,t}$ (standardized)			−0.025			−0.025
			(0.037)			(0.027)
R^2	0.269	0.269	0.269	0.251	0.251	0.251
N	147,228	147,228	147,228	176,173	176,173	176,173

Table VII, part 1. Falsification with nonpolitical and overall risk for investment and employment.

6.6 Table VII: Falsification Exercise: Political Risk, Nonpolitical Risk, and Overall Risk; Table VIII: Variance Decomposition of $\mathrm{PRisk}_{i,t}$

Read: Table VII compares political risk to nonpolitical and overall risk. Table VIII decomposes variation in PRisk using fixed effects.

Detailed interpretation: Table VII shows that nonpolitical risk behaves like uncertainty for investment and employment, but political risk has a stronger connection to lobbying and donations. This helps separate generic uncertainty from political uncertainty. Table VIII is the paper's major stylized fact: time fixed effects explain less than 1 percent of variation, sector-by-time fixed effects explain little relative to total variation, and most variation is firm-level.

Economic message: Political risk is not just a macro time-series. It is granular and firm-specific, which means representative-firm models miss a large part of the action.

TABLE VII

(CONTINUED)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel C	# of recipients$_{i,\,t+1}$			Hedge$_{i,\,t+1}$		
$PRisk_{i,\,t}$ (standardized)	0.491***	0.502***	0.439***	0.007***	0.008***	0.007***
	(0.121)	(0.121)	(0.108)	(0.001)	(0.001)	(0.001)
$NPRisk_{i,\,t}$ (standardized)		−0.042			−0.001	
		(0.052)			(0.001)	
$Risk_{i,\,t}$ (standardized)			0.098			0.001
			(0.101)			(0.002)
R^2	0.148	0.148	0.148	0.141	0.141	0.141
N	176,173	176,173	176,173	176,173	176,173	176,173

Notes. This table explores $PRisk_{it}$'s logical components. $NPRisk_{i,\,t}$ (nonpolitical risk) is calculated in the same way as as $PRisk_{i,\,t}$, but based on nonpolitical bigrams instead of political bigrams. $Risk_{i,\,t}$ counts the number of synonyms of "risk," "risky," "uncertain," or "uncertainty" irrespective of whether they are near a political bigram. As with $PRisk_{i,\,t}$, all measures are relative to the transcript length. The dependent variables are defined as in the preceding tables. Each regression specification controls for $PSentiment_{i,\,t}$, the log of firm assets, as well as time and sector fixed effects. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table VII, part 2. Falsification for political-action outcomes. Political risk, not generic risk, is most relevant for political engagement.

TABLE VIII

VARIANCE DECOMPOSITION OF $PRisk_{it}$

Sector granularity	2-digit SIC (1)	3-digit SIC (2)	4-digit SIC (3)
Time FE	0.81%	0.81%	0.81%
Sector FE	4.38%	6.31%	6.87%
Sector × time FE	3.12%	9.95%	13.99%
"Firm-level"	91.69%	82.93%	78.33%
Permanent differences across firms within sectors (Firm FE)	19.87%	17.52%	16.82%
Variation over time in identity of firms within sectors most affected by political risk (residual)	71.82%	65.41%	61.51%

Table VIII. Variance decomposition. Most variation in PRisk *remains at the firm level even after aggregate and sector effects.*

6.7 Figure IV: Associations between $\text{PRisk}_{i,t}$ and Corporate Actions; Table IX: The Nature of Firm-Level Political Risk

Read: Figure IV visualizes the relation between PRisk and investment, capex guidance, and employment under increasingly demanding fixed effects. Table IX asks whether firm-level political risk is just heterogeneous exposure to aggregate risk.

Detailed interpretation: Figure IV makes the regression patterns visually transparent: the negative relationships survive sector-time and firm fixed effects. Table IX shows that firm-level political risk is not explained by stable exposure to aggregate political risk. Even after beta-style exposure controls, PRisk remains related to volatility and actions.

Economic message: The paper's key empirical claim is that idiosyncratic political risk has economic content. Figure IV and Table IX are designed to prove that this is not merely an aggregate-risk-beta story.

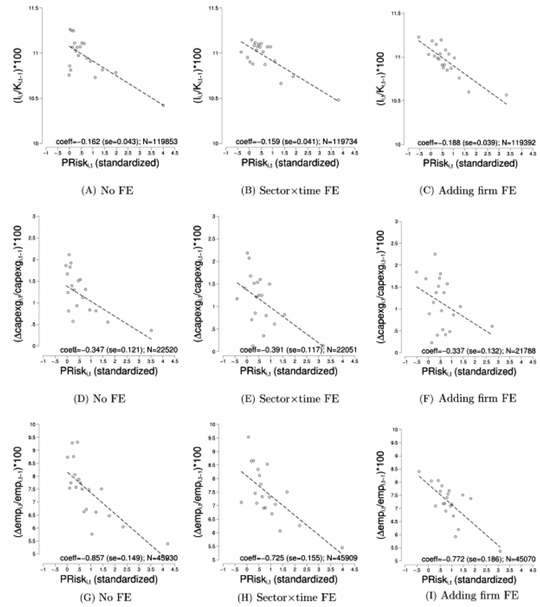

FIGURE IV

Associations between $PRisk_{i,t}$ and Corporate Actions

Figure IV. Added-variable plots linking political risk to investment, capex guidance, and employment.

TABLE IX

THE NATURE OF FIRM-LEVEL POLITICAL RISK

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Implied volatility$_{i,t}$ (standardized)							
$PRisk_{i,t}$ (std.)	0.027***	0.026***	0.026***	0.027***	0.027***	0.029***	0.029***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
β_i × mean of $PRisk_{i,t}$ (std.)		0.001					
		(0.003)					
$\beta_{i,t}$ (2-year rolling) × mean of $PRisk_{i,t}$ (std.)			−0.000				
			(0.000)				
EPU beta$_i$ × mean of $PRisk_{i,t}$ (std.)				0.414			
				(4.764)			
EPU beta (2-year rolling)$_{i,t}$ × mean of $PRisk_{i,t}$ (std.)					0.017		
					(0.063)		
Log(1+$ federal contracts$_{i,t}$)						−0.013***	−0.006
						(0.001)	(0.005)
Log(1+$ federal contracts$_{i,t}$) × mean of $PRisk_{i,t}$ (std.)							−0.001
							(0.001)
R^2	0.501	0.502	0.500	0.501	0.501	0.506	0.506
N	115,059	114,999	110,164	114,979	114,617	115,059	115,059

Table IX, part 1. Tests of whether firm-level political risk is simply exposure to aggregate political risk.

TABLE IX

(CONTINUED)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel B: Realized volatility$_{i,t}$ (standardized)							
$PRisk_{i,t}$ (std.)	0.020***	0.019***	0.020***	0.020***	0.020***	0.021***	0.021***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
β_i × mean of $PRisk_{i,t}$ (std.)		−0.000					
		(0.000)					
$\beta_{i,t}$ (2-year rolling) × mean of $PRisk_{i,t}$ (std.)			0.000				
			(0.000)				
EPU beta$_i$ × mean of $PRisk_{i,t}$ (std.)				9.464***			
				(1.276)			
EPU beta (2-year rolling)$_{i,t}$ × mean of $PRisk_{i,t}$ (std.)					−0.163***		
					(0.014)		
Log(1+$ federal contracts$_{i,t}$)						−0.010***	0.003
						(0.001)	(0.004)
Log(1+$ federal contracts$_{i,t}$) × mean of $PRisk_{i,t}$ (std.)							−0.002***
							(0.001)
R^2	0.490	0.490	0.495	0.490	0.489	0.492	0.493
N	162,153	161,884	153,003	162,153	160,516	162,153	162,153
Time FE	yes	yes	yes	yes	yes	yes	yes
Sector FE	yes	yes	yes	yes	yes	yes	yes
Sector×time FE	yes	yes	yes	yes	yes	yes	yes

Notes. This table is similar to Table IV. It shows results of regressions with realized and implied volatility as the dependent variable in Panels A and B, respectively. β_i is constructed for each firm by regressing $PRisk_{it}$ on its quarterly mean across firms. EPU beta$_i$ is an alternative firm-specific beta obtained from a regression of the firm's daily stock returns on Baker, Bloom, and Davis's (2016) daily Economic Policy Uncertainty (EPU) Index; rolling betas are constructed by running these regressions using observations only from the eight quarters prior to the quarter at hand; mean of $PRisk_{i,t}$ is the cross-sectional average of $PRisk_{i,t}$ at each point in time (standardized by its standard deviation in the time series); and log(1+$ federal contracts$_{i,t}$) is the total amount of federal contracts awarded to firm i in quarter t. All regressions control for the log of firm assets. The dependent variables are defined as in Table IV. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table IX, part 2. Continued tests of the nature of firm-level political risk.

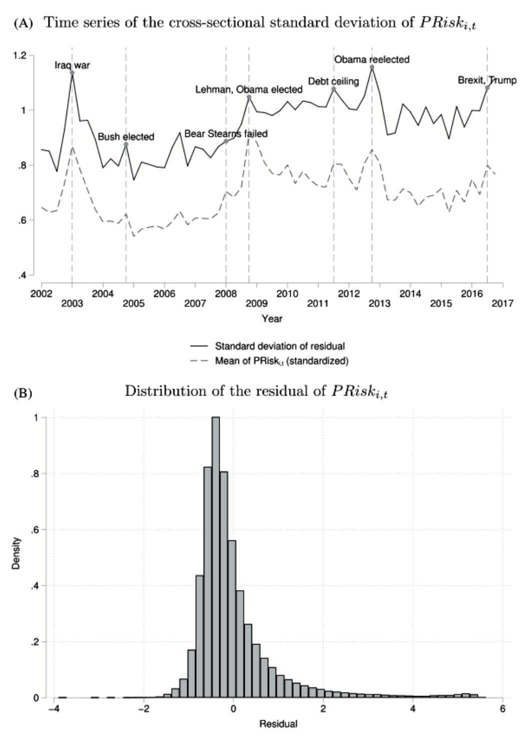

FIGURE V

Figure V. Dispersion of firm-level political risk. Dispersion rises with aggregate political risk and the residual distribution has a fat right tail.

6.8 Figure V: Dispersion of Firm-Level Political Risk; Figure VI: Case Studies; Table X: Topic-Specific Lobbying and Topic-Specific Political Risk

Read: Figure V shows time variation in dispersion and the residual distribution. Figure VI gives two firm case studies. Table X connects topic-specific political risk to topic-specific lobbying.

Detailed interpretation: Figure V shows that when aggregate political risk is high, the cross-sectional dispersion of firm-level political risk also rises. This indicates that political events do not hit firms uniformly. Figure VI shows that peaks in PRisk correspond to concrete firm-specific political episodes. Table X is the strongest topic-level validation: firms that discuss a specific political-risk topic are more likely to lobby on that same topic in the following quarter.

Economic message: Political risk is granular in two ways: across firms and across topics. The paper's measure captures both dimensions, allowing future work to study which political issues matter for which firms.

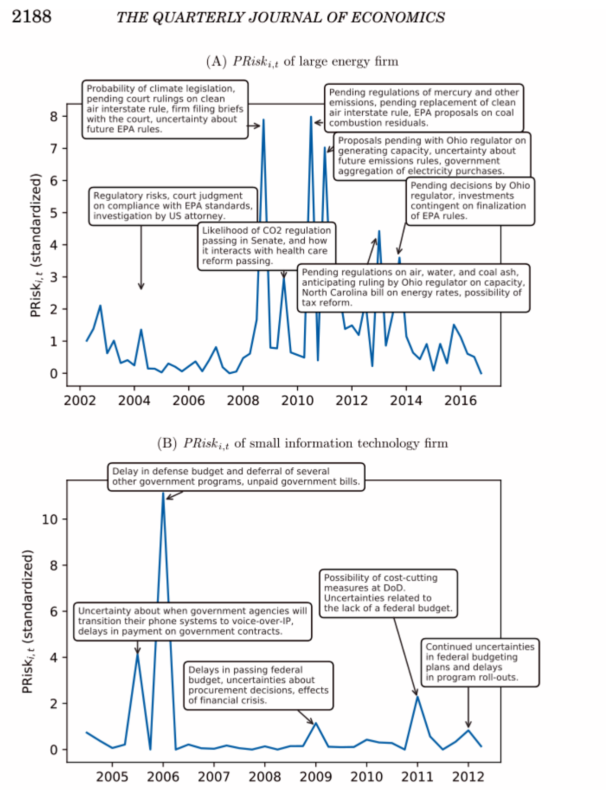

FIGURE VI

Figure VI. Case studies of two firms. The bubbles summarize the political risks driving peaks in the firm-specific series.

TABLE X
TOPIC-SPECIFIC LOBBYING AND TOPIC-SPECIFIC POLITICAL RISK

	(1)	(2)	(3)	(4)	(5)
Panel A: $\mathbb{1}[\text{lobbying}^T_{i,t+1} > 0] * 100$					
$PRisk^T_{i,t}$ (standardized)	1.350***	1.050***	0.794***	0.819***	0.114***
	(0.094)	(0.093)	(0.047)	(0.048)	(0.029)
R^2	0.105	0.127	0.311	0.316	0.647
N	1,177,824	1,177,824	1,177,824	1,177,824	1,177,824
Panel B: Log(1+$ $\text{lobby}^T_{i,t+1}$)					
$PRisk^T_{i,t}$ (standardized)	0.169***	0.133***	0.098***	0.101***	0.015***
	(0.013)	(0.013)	(0.006)	(0.006)	(0.004)
R^2	0.119	0.141	0.352	0.357	0.679
N	1,177,824	1,177,824	1,177,824	1,177,824	1,177,824
Time FE	yes	yes	yes	yes	yes
Sector FE	yes	yes	n/a	n/a	n/a
Topic FE	no	yes	yes	yes	yes
Firm FE	no	no	yes	yes	yes
Sector×time FE	no	no	no	yes	yes
Firm×topic FE	no	no	no	no	yes

Notes. This table shows the results from regressions of a dummy variable that equals 1 if firm i lobbies on topic T in quarter t+1 (Panel A) and the log of 1 plus the firm's lobbying expenditure on topic T in quarter t+1 (Panel B) on the firm's topic-specific political risk in quarter t. The dependent variable in Panel B is calculated under the assumption that firms spread their lobbying expenditure evenly across all topics on which they lobby in a given quarter. Because the lobbying data are semi-annual rather than quarterly before 2007, we drop the first and third quarters prior to 2007 from the sample and assign the outcome variable for the first half of the year to the second quarter and to the fourth quarter for the second half of the year. $PRisk^T_{i,t}$ is standardized by its standard deviation. All specifications control for the log of firm assets. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table X. Topic-specific lobbying and topic-specific political risk. Same-topic lobbying rises after same-topic risk discussion.

expenditures before as much as after the discussion of the political topic at hand.

To probe this possibility, Online Appendix Table XXI replicates Table X, column (5)—our most demanding specification relating lagged $PRisk^T_{it}$ to lobbying at $t+1$—while adding contemporaneous and future $PRisk^T$ to the regression. The results show that the coefficient on the lag is almost unchanged (0.081, std. err. = 0.030), and it shows a larger effect than both the contemporaneous $PRisk^T_{i,t+1}$ (0.064, std. err. = 0.030) and the lead (0.048, std. err. = 0.031), which is statistically indistinguishable from 0. If anything, the lag thus dominates the lead, consistent with a causal interpretation of the results. We interpret this result with caution given the relatively low frequency of the data, the high persistence of lobbying activities,[37] and the fact that the three point estimates are not dramatically different from each other.

The second challenge to a causal interpretation is that a politically engaged firm may lobby the government on a given topic—regardless of the risks associated with the issue—and then have to defend financial or other risks resulting from this lobbying activity during a conference call, or it might lobby in anticipation of future innovations to political risk. Again, the timing of the effect

Table X notes and continuation. The topic-specific design maps conference-call risk discussion to CRP lobbying topics.

7 Short summary

- The paper creates a firm-quarter measure of political risk using earnings-call transcripts.
- PRisk is constructed from political bigrams appearing near risk and uncertainty language, normalized by call length.
- The measure passes qualitative, aggregate, sector, volatility, and political-action validation tests.
- High PRisk predicts higher implied and realized volatility.
- High PRisk predicts lower investment, lower capex guidance, and lower employment growth.
- High PRisk predicts higher lobbying, donations, and politically hedged donations.
- The results survive controls for political sentiment, generic sentiment, stock returns, and earnings surprises.
- Nonpolitical risk predicts generic uncertainty responses, but political risk is more informative about political actions.
- Most variation in the measure is firm-level rather than aggregate or sectoral.
- Cross-sectional dispersion of firm-level political risk rises when aggregate political risk is high.
- Topic-specific risk predicts topic-specific lobbying, validating the topic decomposition.
- The paper's closest causal exercise uses Obama-era budget-crisis terms as instruments for economic-policy political risk.

8 Conclusion

8.1 Core conclusion

Hassan, Hollander, van Lent, and Tahoun create a new measurement infrastructure for political risk. Their central conclusion is that political risk is not merely an aggregate time-series shock. It is also a firm-level, topic-specific, and economically meaningful source of risk that affects volatility, investment, hiring, and political engagement.

8.2 Economic intuition

Firms face political decisions on regulation, taxation, spending, procurement, trade, and enforcement. These decisions can create uncertainty about future cash flows. When the uncertainty is high, firms delay irreversible investments and hiring. At the same time, firms with enough resources may try to manage the risk through lobbying, donations, and political connections.

8.3 Incremental insight relative to prior literature

Prior uncertainty measures often ask whether the economy as a whole is politically uncertain. This paper asks which firms are politically exposed, which political topics matter to them, and what they do in response. The shift from aggregate measurement to firm-level measurement is the key advance.

8.4 Why the firm-level result matters

If political risk differs sharply across firms inside the same sector, then political shocks can distort the allocation of capital and labor across firms. This introduces a misallocation channel beyond the standard aggregate investment channel. The paper therefore connects political risk to granular macroeconomics and productivity losses.

8.5 How to use the measure in future research

The measure can be merged with natural experiments or policy shocks to estimate causal effects of political risk. It can also be decomposed by topic to study which policy domains matter for which firms. The authors provide a template for using text data to build high-frequency firm-level risk measures.

8.6 Limitations

The paper does not claim that every association is causal. Firms may discuss political risk because they anticipate changes in fundamentals. Measurement error is also inevitable in text classification. The paper addresses these issues with multiple validation and falsification exercises, but its broad contribution is measurement and disciplined evidence rather than a single clean experiment.

8.7 Takeaway

Political risk is measurable from firm communication, economically important for firm behavior, and far more heterogeneous than aggregate indices imply. The paper changes the empirical frontier by making political risk observable at the firm-topic-quarter level.